

Three

Address



Code

~...

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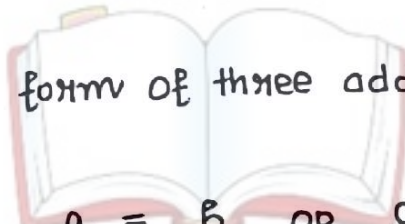


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In three address code, at most three addresses are used to represent any statement.

General form:

The general form of three address code is-


$$a = b \text{ op } c$$

where-

- a, b, c are the operands that can be names, constants or compiler generated temporaries
- op represents the operator

Example :

i> $a = b + c$

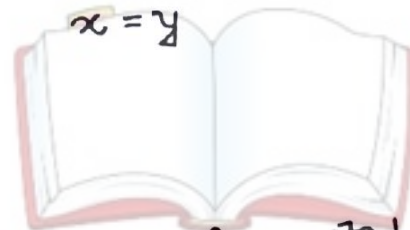
ii> $c = a * b$

Common Three Address Instruction Forms

① Assignment statements :-

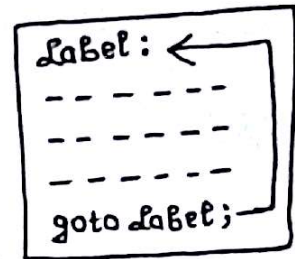
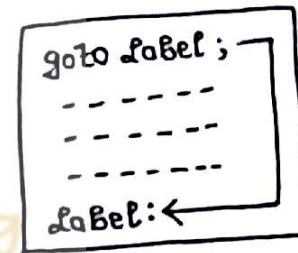
- $x = y \text{ op } z$
- $x = \text{op } y$

② Copy statement :-



③ Conditional jump :-

if $x \text{ rel op } y$ goto L

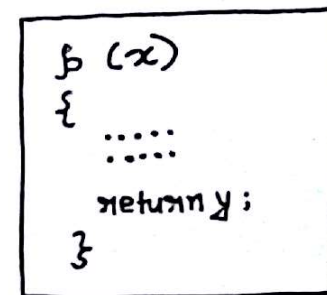


④ Unconditional jump :-

goto L

⑤ Procedure call :-

param x call p
return y

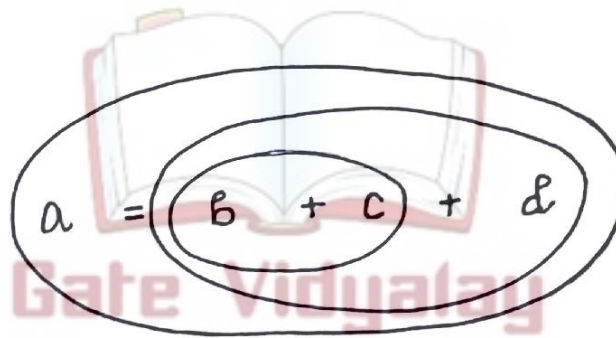


Generation of 3 address code

Illustration-1: Generate 3 address code for -

$$a = b + c + d$$

Soln.:



Three address code will be -

$$T_1 = b + c$$

$$T_2 = T_1 + d$$

$$a = T_2$$

- '+' operator has higher precedence over '=' operator
- '+' operator is left associative.

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Illustration-2: Generate 3 address code for -

$$-(a * b) + (c + d) - (a + b + c + d)$$

Unary operators have higher precedence over binary operators.

Soln:

Three address code will be -

$$\begin{aligned} t1 &= a * b \\ t2 &= \text{uminus } t1 \\ t3 &= c + d \end{aligned}$$

$$t4 = t1 + t3$$

$$t5 = a + b$$

$$t6 = t3 + t5$$

$$t7 = t4 - t6$$

$$- [T1] + [T2] - [T3]$$

$$[T4] + [T2] - [T3]$$

$$[T5] - [T3]$$

$$[T6]$$

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Illustration-3: Generate 3 address code for -

if $A < B$

then 1

else 0

Soln:
2

(1) if $(A < B)$ goto (4)

(2) $T1 = 0$

(3) goto (5)

(4) $T1 = 1$

(5)

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Illustration - 4: _z Generate 3 address code for -

" If $a < b$ and $c < d$ then $t = 1$ else $t = 0$ "

Soln: _z

(1) If $(a < b)$ goto (3)

(2) goto (4)

(3) If $(c < d)$ goto (6)

(4) $t = 0$

(5) goto (7)

(6) $t = 1$

(7)

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